

Quiz 5, DATA 442/642

1. What is the primary goal of the k-means algorithm for data clustering?

- A) Minimizing the variance within each cluster to ensure tighter groupings.
- B) Maximizing the separation between clusters to enhance distinguishability.
- C) Ensuring clusters have the highest possible silhouette score.
- D) Limiting the number of clusters to minimize processing time.

2. What are the common limitations of the k-means algorithm?

- A) Vulnerability to an increased number of features.
- B) Reliance on the initial distribution of data points.
- C) Dependence on initial cluster centers and data assumptions.
- D) Tendency to inaccurately model outliers in the data.

3. In which of the following applications is k-means clustering commonly used?

- A) Segmenting images based on color or texture.
- B) Classifying text documents into predefined categories.
- C) Recognizing spoken words in audio data.
- D) Analyzing patterns in time series data.

4. What is Gaussian Mixture Modeling (GMM), and how does it model data distribution?

- A) GMM models data as a mix of multiple Gaussian distributions.
- B) GMM approximates data distribution using a straight-line fit.
- C) GMM creates branches of decisions based on data characteristics.
- D) GMM uses neural networks to model complex patterns in data.

5.. Which of the following tasks can benefit from using Gaussian Mixture Models (GMMs)?

- A) Detecting anomalies in data patterns.
- B) Compressing images by reducing redundant information.
- C) Reducing the dimensionality of feature spaces.
- D) Modeling relationships between dependent and independent variables.

6. How are Gaussian Mixture Models (GMMs) utilized for anomaly detection?

- A) By identifying dense regions within clusters.
- B) By identifying data points that don't align with the expected distribution.
- C) By eliminating data points considered as outliers.
- D) By calculating a regression fit to identify anomalies.

7. Why is it important to choose an appropriate number of clusters in Gaussian Mixture Models (GMMs)?

- A) To lower the computational costs associated with the model.
- B) To make the clustering results more understandable.
- C) To improve the accuracy and relevance of clustering results.
- D) To prevent the model from fitting irrelevant details in the data.

8. What are some techniques for selecting the optimal number of clusters in Gaussian Mixture Models (GMMs)?

- A) The Elbow Method, which visualizes variance reduction.
- B) Silhouette analysis, which assesses cluster cohesion and separation.
- C) Hierarchical clustering, which creates a tree of clusters.
- D) Random initialization to observe clustering outcomes.

9. Which of the following methods evaluates both the compactness of clusters and the degree of separation between them to determine the optimal number of clusters?

- A) The Elbow Method, which focuses on within-cluster variance reduction.
- B) Silhouette analysis, assessing cohesion and distance between clusters.
- C) Hierarchical clustering, creating nested clusters at multiple levels.
- D) K-means clustering, using iterative optimization for cluster assignments.

10. Given three 2D points: $A(0,0)$, $B(1,0)$, and $C(2,0)$, and the following Similarity measure:

$$w_{ij} = \begin{cases} 1 & \text{if } d(x_i, x_j) \leq 1 \\ 0 & \text{otherwise,} \end{cases}$$

where $d(x_i, x_j)$ is the Euclidean distance, compute the Laplacian matrix.